The Gambler's Ruin Problem for One-Dimensional Random Walks: Simple Symmetric Random Walk and Some Extensions

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December 22, 2025

Abstract

The gambler's ruin problem serves as a classical entry point into random walk theory. For the simple symmetric random walk, explicit expressions for hitting probabilities can be derived using difference equations or martingale methods. However, for more general step distributions, the analysis becomes challenging due to the "overshoot" phenomenon. This study explores the problem in three settings: the simple symmetric random walk, the spread-out model where steps are bounded, and the finite-variance case. The methods rest on martingale theory, in particular the optional sampling theorem, whose role we make explicit throughout. This report is expository in nature: all results are drawn from the cited literature, and our aim is a unified, self-contained account rather than the establishment of new theorems.

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1 Introduction

We begin by introducing the fundamental concepts of our model, starting with the random walk, the stopping time, and the formulation of the gambler's ruin problem. This work is a study report rather than original research: the statements presented are established results, which we follow from Lawler and Limic (2010) and Koralov and Sinai (2007) rather than claim as new, our contribution being a unified and self-contained exposition across the three settings. Unless otherwise stated, the definitions and theorems follow Lawler and Limic (2010).

1.1 Random Walk and Stopping Time

A random walk is defined as a sequence $S = (S_0, S_1, S_2, \ldots)$, where S_n is the sum of the starting point S_0 and independent, identically distributed steps X_i such that

$$S_n = S_0 + X_1 + \cdots + X_n.$$

In this work, we specifically focus on the unbiased walk where the expectation is zero, $\mathbb{E}(X_i) = 0$.

Next, we define the stopping time η_r as the first time n such that the random walk S exits the interval $(0, r)$. This can be written as

$$\eta_r = \inf\{n \in \mathbb{N} : S_n \leq 0 \text{ or } S_n \geq r\}.$$

For a simple symmetric random walk, the stopping time is finite almost surely, meaning that $\mathbb{P}(\eta_r < \infty) = 1$. This can be understood intuitively through a block argument. In any block of r steps, there is a strictly positive probability $\varepsilon \geq (1/2)^r$ that the walk moves r steps to the right consecutively, guaranteeing an exit eventually.

1.2 The Gambler's Ruin Problem

The problem is formulated as follows. A gambler starts to bet with initial wealth $S_0 = x \in (0, r)$. We are interested in the probability of winning, which means reaching the intended wealth r before going broke at 0. Mathematically, we want to evaluate

$$\mathbb{P}^x\left(S_{\eta_r} \geq r\right),$$

i.e., the probability that the ending position is at least the intended wealth r starting from the initial position x.

2 The Simple Symmetric Random Walk

For a simple random walk, the step probabilities are defined as $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = -1) = q = 1 - p$. In the symmetric case, we have $p = q = \frac{1}{2}$. Assuming x and r are integers, the problem simplifies to finding $\mathbb{P}^x\left(S_{\eta_r} = r\right)$.

It is worth noting that if the walk were biased with $p \neq \frac{1}{2}$, the linear scaling we are about to derive would break down. Solving the same recurrence in the biased case gives

$$\mathbb{P}^x\left(S_{\eta_r} = r\right) = \frac{1 - (q/p)^x}{1 - (q/p)^r},$$

so the winning probability depends exponentially on x and r through the ratio q/p rather than scaling linearly; for instance, when $p > \frac{1}{2}$ with x fixed, it converges to $1 - (q/p)^x > 0$ as $r \to \infty$. Since we assume symmetry, however, we may use the following two approaches.

2.1 Difference Equations Approach

This method follows Koralov and Sinai (2007). By conditioning on the first step, we obtain the relation

$$\begin{aligned}\mathbb{P}^x\left(S_{\eta_r} = r\right) &= \mathbb{P}^x\left(S_{\eta_r} = r \,\middle|\, X_1 = 1\right) \cdot \mathbb{P}(X_1 = 1) + \mathbb{P}^x\left(S_{\eta_r} = r \,\middle|\, X_1 = -1\right) \cdot \mathbb{P}(X_1 = -1) \\ &= \mathbb{P}^{x+1}\left(S_{\eta_r} = r\right) \cdot p + \mathbb{P}^{x-1}\left(S_{\eta_r} = r\right) \cdot (1 - p).\end{aligned}$$

Writing $f(x) = \mathbb{P}^x\left(S_{\eta_r} = r\right)$, the symmetric case $p = 1/2$ reduces this relation to the harmonic recurrence

$$f(x) = \frac{1}{2} \cdot f(x+1) + \frac{1}{2} \cdot f(x-1),$$

with boundary conditions $f(0) = 0$ and $f(r) = 1$ for $x \in \{1, \ldots, r-1\}$. The unique solution to this linear recurrence is

$$f(x) = \frac{x}{r}.$$

We can extend this model with the same philosophy to a countable-step-range case or a continuous-step-range case. For the countable case, the relation becomes

$$\begin{aligned}
\mathbb{P}^x\left(S_{\eta_r} = r\right) &= \sum_k \mathbb{P}^x\left(S_{\eta_r} = r \,\middle|\, X_1 = k\right) \cdot \mathbb{P}(X_1 = k) \\
&= \sum_k \mathbb{P}^{x+k}\left(S_{\eta_r} = r\right) \cdot \mathbb{P}(X_1 = k),
\end{aligned}$$

while for the continuous case, we have

$$\begin{aligned}
\mathbb{P}^x\left(S_{\eta_r} \geq r\right) &= \int_{\mathbb{R}} \mathbb{P}^x\left(S_{\eta_r} \geq r \,\middle|\, X_1 = y\right) f_{X_1}(y)\,\mathrm{d}y \\
&= \int_{\mathbb{R}} \mathbb{P}^{x+y}\left(S_{\eta_r} \geq r\right) f_{X_1}(y)\,\mathrm{d}y \\
&= \int_{\mathbb{R}} \mathbb{P}^{x+y}\left(S_{\eta_r} \geq r\right) \mathbb{P}(X_1 \in \mathrm{d}y).
\end{aligned}$$

For the integrand to be defined on all of $\mathbb{R}$, we read $\mathbb{P}^z\left(S_{\eta_r} \geq r\right)$ for a starting point z outside $(0, r)$ by the convention $\mathbb{P}^z\left(S_{\eta_r} \geq r\right) = 1$ when $z \geq r$ and $= 0$ when $z \leq 0$, which is consistent with $\eta_r = 0$ in those cases. This makes the relation meaningful even when the first step y already carries the walk past a boundary.

2.2 Martingale Approach

Alternatively, we can employ martingale theory. Let $M_n = S_{n \wedge \eta_r}$. The stopped process M_n is a bounded martingale. By the optional sampling theorem, we have $\mathbb{E}^x(M_0) = \mathbb{E}^x\left(M_{\eta_r}\right)$. This implies that

$$x = r \cdot \mathbb{P}^x\left(S_{\eta_r} = r\right) + 0 \cdot \mathbb{P}^x\left(S_{\eta_r} = 0\right),$$

which yields the same result

$$\mathbb{P}^x\left(S_{\eta_r} = r\right) = \frac{x}{r}.$$

3 The Spread-Out Model (Bounded Steps)

In the previous section, the analysis relied on the fact that the walk stops exactly at 0 or r. When the step distribution allows values other than 1 and -1, specifically continuous random variables or larger discrete jumps, the walk will likely overshoot the boundaries. This phenomenon complicates the exact calculation of the hitting probability.

We now assume that the steps X_i are independent, identically distributed random variables with mean zero. We further assume the steps are bounded by K almost surely and satisfy a non-degeneracy condition.

3.1 The Overshoot Phenomenon

The stopping time η_r remains the first time the walk exits $(0, r)$. However, at time η_r, the position S_{η_r} is not necessarily 0 or r. Instead, if the gambler wins, the position lands in the interval $[r, r + K]$, and if the gambler loses, it lands in $[-K, 0]$.

Because of this, the relation derived from the optional sampling theorem, $x = \mathbb{E}^x\left(S_{\eta_r}\right)$, no longer simplifies to $x = r \cdot \mathbb{P}\left(S_{\eta_r} = r\right)$.

3.2 Deriving the Bounds via Martingale

Despite the overshoot, the martingale property provides useful bounds. If we assume the steps are bounded by some constant K, so that $\mathbb{P}(|X_i| > K) = 0$, the stopped process $S_{n \wedge \eta_r}$ is uniformly bounded in $[-K, r + K]$. Thus, the optional sampling theorem holds and gives

$$x = \mathbb{E}^x\left(S_{\eta_r}\right).$$

We can decompose this expectation based on the winning and losing events. Let $u(x)$ denote $\mathbb{P}^x\left(S_{\eta_r} \geq r\right)$ for shorthand. We then have

$$x = \mathbb{E}^x\left(S_{\eta_r} \mid S_{\eta_r} \geq r\right)u(x) + \mathbb{E}^x\left(S_{\eta_r} \mid S_{\eta_r} \leq 0\right)(1 - u(x)).$$

Using the bounds of the overshoot, we can establish the inequality

$$r \cdot u(x) - K \cdot (1 - u(x)) \leq x \leq (r + K) \cdot u(x) + 0.$$

Rearranging these terms gives us the basic bounds

$$\frac{x}{r + K} \leq \mathbb{P}^x\left(S_{\eta_r} \geq r\right) \leq \frac{x + K}{r + K}.$$

3.3 An Estimate Theorem

While the basic bounds are useful, another estimate exists that preserves the linear scaling behavior seen in the simple symmetric case; we will see this form later in Section 4.

The following estimate follows Lawler and Limic (2010, 103–112). Let $K < \infty$. For a one-dimensional random walk, if the steps satisfy the boundedness condition $\mathbb{P}(|X_i| > K) = 0$, the mean zero condition $\mathbb{E}(X_i) = 0$, and the non-degeneracy condition $\mathbb{P}(X_i > 0) > 0$, then there exist constants c_1 and c_2, depending on K, such that

$$c_1 \frac{x + 1}{r} \leq \mathbb{P}^x\left(S_{\eta_r} \geq r\right) \leq c_2 \frac{x + 1}{r}.$$

This confirms that even with the “fuzziness” of the boundary due to overshoot, the probability of winning still scales linearly with the initial position x relative to the intended wealth r.

The appearance of $x + 1$ rather than x in the numerator is deliberate. For $x \geq 1$ the two differ only by a constant factor, but as $x \to 0$ the $+1$ records an *entrance effect:* even a walk starting arbitrarily close to 0 can still win with probability of order $1/r$ by taking a single sufficiently large first step, a possibility guaranteed by the non-degeneracy condition. This is precisely where the estimate sharpens the basic bounds of Section 3. The upper bound $(x + K)/(r + K)$ is already of order $(x + 1)/r$, so the estimate adds nothing there; the gain is in the lower bound, since $x/(r + K)$ degenerates to 0 as $x \to 0$ whereas the estimate keeps it at order $1/r$. The same linear form, obtained here for bounded steps, will reappear in Section 4 under the weaker finite-variance assumption.

4 Finite-Variance Case

The previous model assumed that the step size was strictly bounded. However, in many applications, steps may follow distributions with infinite support, such as a normal distribution. A natural question arises regarding whether the linear scaling of the winning probability is preserved if we relax the boundedness condition.

The answer is affirmative, provided the random walk has a finite variance.

4.1 The Uniform Estimate Theorem

We consider a random walk where the steps X_i are not necessarily bounded but satisfy specific regularity conditions. The strength of this result is that the constants in the estimate depend only on a set of family parameters: K, δ, b, and ρ.

Before stating the theorem, let us interpret the physical meaning of these parameters. First, we redefine K; it is now a variance bound, i.e.,

$$\mathbb{E}\left(X_i^2\right) \leq K^2.$$

Second, δ is a non-degeneracy parameter; it ensures that the walk is not stagnant and has a minimum probability of moving significantly to the right,

$$\mathbb{P}(X_i \geq 1) \geq \delta.$$

Third, b represents a local survival probability, guaranteeing that the walk has a chance to stay above a certain negative threshold for a block of time,

$$\inf_{n \in \mathbb{N}} \mathbb{P}\left(\bigcap_{i=1}^{n^2} S_i > -n\right) \geq b.$$

Finally, ρ controls the global fluctuation, ensuring the walk has enough variance to eventually drop below a certain level,

$$\inf_{n \in \mathbb{N}} \mathbb{P}\left(S_{n^2} \leq -n\right) \geq \rho.$$

We thus present the general estimation result, often referred to as the gambler's ruin estimate, following Lawler and Limic (2010, 103–112). Let $\delta, K \in \mathbb{R}^+$ and let $b, \rho \in (0,1)$. Let S be a random walk satisfying the four conditions described above. Then, there exist constants c_1 and c_2 dependent only on these parameters such that

$$c_1 \frac{x+1}{r} \leq \mathbb{P}^x\left(S_{\eta_r} \geq r\right) \leq c_2 \frac{x+1}{r}.$$

It is crucial to note that the finite variance condition (K) is necessary here. To see that variance is the binding constraint, the counterexample must still respect our standing assumption $\mathbb{E}(X_i) = 0$. A Cauchy distribution does not: it has no finite first moment, so $\mathbb{E}(X_i)$ is undefined and S_n is not even a martingale; it would confound the failure of integrability with that of finite variance rather than isolate the latter. A clean example is instead a symmetric distribution with heavy tails $\mathbb{P}(|X_i| > t) \sim t^{-\alpha}$ as $t \to \infty$ for some $1 < \alpha < 2$. Here $\mathbb{E}|X_i| < \infty$, so the mean-zero assumption holds and S_n remains a genuine martingale, yet $\mathbb{E}\left(X_i^2\right) = \infty$. With such steps the walk can make a single large jump from deep inside the interval to far outside, bypassing the boundary entirely, and the probability of winning no longer scales linearly as x/r. The precise scaling in this regime is governed by the index α and is a separate, known result.

4.2 Sketch of the Derivation

The proof is significantly more involved than the bounded case because the overshoot $S_{\eta_r} - r$ can be very large. We cannot simply bound the position by $r + K$.

For the upper bound estimate, we rely on controlling the overshoot. We utilize the overshoot lemma, which provides estimates for the probability of the walk going very fast. The derivation involves constructing constants based on the family parameters, ultimately leading to an upper bound coefficient proportional to the variance bound.

For the lower bound estimate, we return to the martingale property $x = \mathbb{E}^x\left(S_{\eta_r}\right)$. This identity now requires a different justification than in Section 3: the steps are unbounded, so the stopped process

$S_{n\wedge\eta_r}$ is no longer confined to $[-K, r+K]$ and the uniform-boundedness argument used there no longer applies. Instead, the finite-variance and non-degeneracy conditions guarantee $\mathbb{E}(\eta_r) < \infty$, and Wald's second identity gives

$$\mathbb{E}^x\left[\left(S_{n\wedge\eta_r} - x\right)^2\right] = \sigma^2\,\mathbb{E}^x[n\wedge\eta_r] \le \sigma^2\,\mathbb{E}^x(\eta_r) < \infty,$$

where $\sigma^2 = \mathbb{E}\left(X_i^2\right) \le K^2$. Hence $\left(S_{n\wedge\eta_r}\right)_{n\in\mathbb{N}}$ is bounded in L^2, therefore uniformly integrable, and the optional sampling theorem applies, yielding $x = \mathbb{E}^x\left(S_{\eta_r}\right)$ as claimed. We then decompose the expectation based on the magnitude of the win into three parts: losses ($S_{\eta_r} \le 0$), moderate wins ($r \le S_{\eta_r} \le (1+t_0)r$), and huge wins ($S_{\eta_r} \ge (1+t_0)r$).

Since the loss term is non-positive, we focus on the wins. By choosing a threshold parameter t_0 sufficiently large, we can bound the contribution of huge wins such that

$$\mathbb{E}^x\left(S_{\eta_r}; S_{\eta_r} \ge (1+t_0)r\right) \le \frac{x}{2}.$$

This forces the moderate wins to carry significant weight, implying that

$$\mathbb{E}^x\left(S_{\eta_r}; r \le S_{\eta_r} \le (1+t_0)r\right) \ge \frac{x}{2}.$$

Since the value of S_{η_r} in this term is at most $(1+t_0)r$, we can deduce that

$$(1+t_0)r \cdot \mathbb{P}^x\left(S_{\eta_r} \ge r\right) \ge \frac{x}{2}.$$

Rearranging this yields the lower bound

$$\mathbb{P}^x\left(S_{\eta_r} \ge r\right) \ge \frac{1}{2(1+t_0)}\frac{x}{r}.$$

This sketch produces the form x/r rather than the $(x+1)/r$ of the theorem; the two agree up to a constant for $x \ge 1$, and the $+1$ reflects the same entrance effect discussed for the bounded estimate in Section 3, which a bare x/r would miss as $x \to 0$.

While the existence of constants c_1 and c_2 is guaranteed, deriving explicit expressions in terms of the family parameters is analytically complex.

5 Conclusion and Some More Interesting Problems

Among all three models, we can see the shadow of the linear scaling $\frac{x}{r}$ existing over each solution or estimate.

The gambler's ruin problem is just the tip of the iceberg in random walk theory. Here we briefly introduce three directions for further exploration, focusing on the mathematical structures that emerge in higher dimensions and continuous limits; standard treatments of these topics can be found in Lawler and Limic (2010).

5.1 Connection to Potential Theory (Two-Dimensional Extension)

One of the most beautiful results in probability theory is the connection between random walks and electrostatics. When we extend the problem to two dimensions, for instance, a random walk on a grid $\mathbb{Z}^2$, the recurrence relation for the probability of winning $u(x, y)$ satisfies the discrete mean value property

$$u(x, y) = \frac{1}{4}(u(x+1, y) + u(x-1, y) + u(x, y+1) + u(x, y-1)).$$

This can be rewritten using the discrete Laplacian operator Δ as

$$\Delta u(x, y) = u(x+1, y) + u(x-1, y) + u(x, y+1) + u(x, y-1) - 4u(x, y) = 0.$$

This is the discrete analogue of Laplace's equation $\Delta V = 0$. In physics, the electric potential in a charge-free region satisfies this property. Thus, calculating the gambler's ruin probability on a grid is mathematically identical to solving for the electric potential on a discrete network of resistors. Theoretically, the shape of the boundary does not change the nature of this connection; the probability is determined by the harmonic measure of the boundary.

5.2 Expected Hitting Time and the Poisson Equation

Besides the probability of winning, another natural question is: how long does the game last? Let $h(x) = \mathbb{E}^{x(\eta_r)}$ be the expected hitting time. For the simple symmetric random walk, we can derive this using a similar difference equation approach. By conditioning on the first step, we have

$$h(x) = 1 + \frac{1}{2} \cdot h(x+1) + \frac{1}{2} \cdot h(x-1).$$

The constant term 1 accounts for the single time step increment at the first transition. Rearranging terms, we see that the second difference is constant,

$$h(x+1) - 2h(x) + h(x-1) = -2.$$

This is the discrete analogue of the Poisson equation $\Delta u = f$, where f is a non-zero constant. With boundary conditions $h(0) = 0$ and $h(r) = 0$ (since the game ends immediately at the boundaries), the solution is given by the parabolic function

$$h(x) = x \cdot (r - x).$$

This reveals a profound link to harmonic analysis: while the hitting probability corresponds to the Laplace equation (harmonic functions), the expected duration corresponds to the Poisson equation.

5.3 Continuous-Time Limit: Brownian Motion

Finally, if we let the step size $\Delta x \to 0$ and the time step $\Delta t \to 0$ in a specific way where $\Delta t \approx (\Delta x)^2$, the discrete random walk converges to a continuous stochastic process known as Brownian motion, denoted by B_t.

In this limit, the discrete difference equation transforms into a differential equation. For the one-dimensional case, the relation $f(x) = \frac{1}{2} \cdot f(x + \Delta x) + \frac{1}{2} \cdot f(x - \Delta x)$ becomes

$$\frac{1}{2} f''(x) = 0.$$

The solution to this ordinary differential equation with boundary conditions $f(0) = 0$ and $f(r) = 1$ is simply the linear function

$$f(x) = \frac{x}{r}.$$

This confirms that our discrete result $\frac{x}{r}$ is consistent with the continuous theory. The overshoot problem disappears in Brownian motion because the path is continuous, ensuring it must hit the boundary exactly without jumping over it.

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